Quality Management

Chapter 1: Introduction to Total Quality Management (TQM) & 5S

A Comprehensive Study Guide

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Prepared for exam revision & deep understanding

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1. Introduction — Why Quality Matters

Quality is not just about making a good product — it touches every aspect of how a business operates, from design to delivery to customer satisfaction. This chapter introduces the foundations of **Total Quality Management (TQM)** and explains why it has become essential for organizations worldwide.

Real-World Examples: Corrigendum Notices

The lecture begins with real-world examples of **corrigendum notices** (official correction notices) from Indian organizations:

- **Mahanagar Telephone Nigam Ltd (MTNL), Delhi:** A tender notice incorrectly stated "ASTT GENERAL MANAGER (MM-III)" when it should have been "ASTT GENERAL MANAGER (M-III)." A corrigendum had to be published to correct this error.
- **Pondicherry University:** A Notice Inviting Tender (NIT) for security services had an error — "Date of opening of financial bids" was listed as "The tenders/short listed in Pre-qualification bid will be informed later" instead of "Will be informed later." Again, a corrigendum was needed.
- **University of Calcutta:** The name of an agency was spelled incorrectly — "Electro Equipments Trading Corporation" was wrongly written as "Electro Ecuidements Trading Corporation."

> **Analysis Point:** These examples show that even small errors in official documents can cause confusion and require costly corrections. Quality isn't just about manufacturing — it starts with getting the basics right in documentation, communication, and processes. If prestigious organizations make such errors, imagine the impact in manufacturing or service delivery where the stakes are even higher.

2. Changes in Indian Business Environment

The Indian business environment has undergone significant transformation, creating both opportunities and challenges for quality management:

- **Increase in the role of private sector and foreign investment** — Liberalization (post-1991) opened India to global competition, forcing companies to improve quality standards.

- **Large domestic market provided by vast and growing middle class population** — A massive consumer base that increasingly demands quality products.
- **Availability of skilled manpower and professional managers at moderate cost** — India has the human resources to implement quality systems, but the mindset needs to change.
- **Well developed capital markets, banking infrastructure and financial service sector** — The financial ecosystem supports investment in quality improvement initiatives.

> **Analysis Point:** While India has all the ingredients for world-class quality (skilled people, large market, capital), the gap lies in quality culture and mindset. The data in the following sections shows that India still lags behind countries like Japan and Singapore in quality practices.

3. Quality Ratings — India vs the World

Three key metrics were compared across 7 countries on a 10-point scale. The data paints a clear picture of where India stands globally in quality.

3.1 Price to Quality Rating (10-Point Scale)

This measures how well a country's products balance price and quality — higher means better value.

Country	Rating	Visual
Japan	**7.03**	
Singapore	**7.00**	
Germany	**6.33**	
USA	**5.92**	
Thailand	**5.67**	
Taiwan	**5.57**	
India	**3.47**	

3.2 Customer Orientation Rating (10-Point Scale)

This measures how well businesses in each country focus on customer needs and satisfaction.

Country	Rating	Visual
Japan	**7.94**	
Singapore	**6.46**	
Germany	**6.45**	
USA	**6.24**	
Thailand	**5.78**	
Taiwan	**5.67**	
India	**3.37**	

3.3 Practice of TQM Rating (10-Point Scale)

This measures the actual implementation and practice of TQM principles in each country.

Country	Rating	Visual
Japan	**8.66**	
Singapore	**7.18**	
Germany	**7.04**	
USA	**6.64**	
Thailand	**5.68**	
Taiwan	**4.80**	
India	**4.06**	

Analysis Point: India ranks **last in all three metrics** among these 7 countries. The gap is especially alarming in Price-to-Quality (3.47 vs Japan's 7.03 — nearly half) and Customer Orientation (3.37 vs Japan's 7.94). Japan leads in all categories, which is not surprising given its long history of quality culture (Toyota Production System, Kaizen, etc.). This data underscores why India urgently needs TQM adoption.

4. World Economic Forum Survey — India's Rankings

The World Economic Forum ranked countries across various factors. India's rankings out of 53 countries reveal a mixed picture:

Factor	India's Rank (out of 53)	Assessment
No. of Scientists and Engineers	**3**	Excellent — India has a massive talent pool
Primary Education (Male)	**8**	Good
Science and Math Education Levels	11	Above average
Indigenous Innovation	35	Below average
Technology Rating	36	Below average
Secondary Education (Male)	39	Poor
Global Technological Leadership	41	Poor
Corporate R&D Spending	44	Very poor
Secondary Education (Female)	47	Very poor
Primary Education (Female)	48	Very poor
Quality of Engineers	**52**	Almost last!

Analysis Point: India has a paradox — it ranks **3rd in the number of scientists and engineers**, but **52nd out of 53 in quality of engineers**. This means India produces a large quantity of engineers, but the quality of their education and skills is severely lacking. Similarly, female education at both primary (48th) and secondary (47th) levels is alarmingly poor. The key takeaway: quantity without quality is meaningless — which is exactly what TQM teaches.

5. Mahatma Gandhi's View on the Customer

"A Customer is the most important visitor on our premises.

He is not dependent on us. We are dependent on him.

He is not an interruption on our work. He is the purpose of it.

He is not an outsider on our business. He is a part of it.

We are not doing him a favor by serving him... He is doing us a favor by giving us an opportunity to do so."

— **Mahatma Gandhi**

Analysis Point: This powerful quote captures the essence of **customer-centric philosophy** that lies at the heart of TQM. Long before modern management theories, Gandhi understood that the customer is not a burden but an opportunity. In TQM, customer satisfaction is the ultimate measure of quality. Every process, every employee, every decision should be oriented towards serving the customer better.

6. Characteristics of Indian Work Profile

Before TQM can be implemented, we must understand the existing workplace culture. Indian organizations often exhibit these characteristics:

Characteristic	What It Means & Why It's a Problem
Negative Perception	Workers tend to see quality initiatives as additional burden rather than improvement opportunity
Lack of Motivation	Low employee engagement leads to poor quality output
No Serious Concern for Training	Organizations don't invest enough in skill development
Individual Guided	Decisions depend on individuals rather than systems — creates inconsistency
Myopic View on ISO 9000	Companies treat ISO certification as a one-time checkbox, not a continuous practice
Poor Understanding of Quality	Quality is seen as "inspection" rather than a comprehensive management philosophy
Reactive Approach	Problems are fixed after they occur, rather than prevented proactively
Low Literacy	Limits the ability to implement documentation-heavy quality systems at shop-floor level

7. Internal & External Environment of Indian Organizations

7.1 Internal Environment Issues

- **Learning Disabilities** — Organizations struggle to learn from mistakes and adapt
- **Employee Attitudes** — Resistance to change and lack of ownership towards quality
- **Competitive Disadvantage** — Without quality, Indian firms lose to global competitors
- **Low Automation** — Manual processes lead to more errors and inconsistency
- **Lack of Emphasis on Research and Development** — R&D investment is too low to drive innovation

7.2 External Environment Factors

- **General Environment** — Political, social, and legal factors that impact business
- **Globalizations and Competition** — International companies bring superior quality, raising the bar
- **Economic Climate** — Economic conditions affect investment in quality programs
- **Extent of Technology Transfer and Use** — Access to advanced technology is crucial for quality improvement

8. SWOT Analysis of Indian Industry

A comprehensive SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis helps understand where Indian industry stands in terms of quality readiness.

STRENGTHS

- Cultural adaptability in the right environment
- Financial participation by foreign agencies
- Customer-driven pressures for quality
- Large industrial base and manpower
- Technical support of collaborators
- Indian culture and value system
- Increased awareness for quality
- Indian tradition of excellence
- Easy access to high technology
- Abundant natural resources
- Large domestic market
- Long employment & cheap labor

WEAKNESSES

- Lack of trust and credibility in the working system
- Lack of clarity/seriousness for achieving targets
- Lack of precise observance of rules and norms
- Lack of knowledge regarding SQC techniques
- Low quality of bought-out items/components
- Lack of consciousness of time as money
- Management views short-term benefits
- Politicalization of labor unions
- Lack of accountability for actions
- Self-before-organization attitude
- Lack of top management commitment
- Inadequate economic resources
- Lack of indigenous technology
- Inadequate infrastructure
- Quantity before quality mindset
- Lack of team spirit & planning
- Cartel formation & seller's market

OPPORTUNITIES

- Up-gradation of manufacturing and test facilities
- Weaknesses as opportunities for improvement
- Improvement in quality of supply chain
- Computerization and automation
- Large manufacturing capacity
- Better vendor development
- Present government policy (supportive)
- Education and training programs
- International market access
- Reduced lead time through technology
- Globalization

THREATS

- Complicated government regulation and red tape
- Passive attitude of industry towards quality
- Lack of stable political and cultural ethos
- Negligible application research
- Negligible industrial training investment
- Lack of professional attitude
- Recession in global economy
- Resistance to change
- Un-preparedness for global competition
- Globalization (both opportunity and threat)

Analysis Point: Notice that **Globalization** appears in both Opportunities and Threats. This is because globalization opens new markets (opportunity), but also brings fierce competition from quality-focused nations (threat). The key differentiator is whether Indian industry proactively embraces quality or remains passive. Also notable: the weaknesses list is significantly longer than strengths, suggesting that India has more ground to cover in quality management.

9. Quality Misconceptions — Common Myths Debunked

Many people hold incorrect beliefs about quality. Let's examine and debunk the five most common misconceptions:

#	Misconception	Reality
1	**Better quality requires higher cost**	Actually, investing in quality reduces total cost by eliminating rework, returns, warranty claims, and waste. Prevention is cheaper than correction.
2	**Quality is innately perceived as separating good from bad**	Quality is not about sorting good from bad after production. It's about designing and building processes so that defects don't occur in the first place.
3	**Increased quality can be achieved through investment in technology**	Technology alone is not enough. Quality requires a combination of proper processes, trained people, management commitment, and cultural change.
4	**Quality is confined to the product and manufacturing process**	Quality applies to every function — marketing, HR, finance, customer service, documentation. It's an organization-wide concept.
5	**Quality is the responsibility of the quality manager and his department**	Quality is everyone's responsibility — from the CEO to the shop floor worker. TQM means "Total" involvement.

10. Quality Characteristics

A **Quality Characteristic** is one or more elements that define the intended quality level of a product or service. These can be grouped into four categories:

Type	Description	Examples
Structural Characteristics	Physical and measurable properties of the product	Length of a part, weight of a can, strength of a beam, viscosity of a fluid
Sensory Characteristics	Properties perceived through human senses	Taste of food, smell of perfume, beauty of a model, appearance of a car
Time-Oriented Characteristics	Properties related to performance over time	Warranty period, reliability (MTBF), maintainability
Ethical Characteristics	Properties related to values and behavior	Honesty, courtesy, friendliness of service staff

Analysis Point: Notice that quality is not just about measurable, physical properties. Sensory and ethical characteristics are equally important, especially in services. A restaurant might have perfectly safe food (structural) but terrible customer service (ethical), which will still result in poor perceived quality.

11. Dimensions of Product Quality — Garvin's 8 Dimensions

David A. Garvin identified **eight dimensions of quality** from a manufacturing perspective. These provide a comprehensive framework for evaluating product quality:

Dimension	Key Question	Explanation
1. Performance	Will the product do the intended job?	The primary operating characteristics. For a car: speed, acceleration, fuel efficiency. For a TV: picture clarity, sound quality.
2. Conformance	Does it meet the specifications?	How well the product matches its design specifications and industry standards. Zero defects is the ideal.
3. Reliability	How often does the product fail?	The probability that the product will function without failure over a specified period. Measured as MTBF (Mean Time Between Failures).
4. Durability	How long does the product last?	The total useful life of the product before it needs to be replaced. A light bulb rated for 10,000 hours has higher durability than one rated for 1,000 hours.
5. Serviceability	How easy is it to repair?	Speed, ease, and cost of repair. A product with good serviceability can be fixed quickly and cheaply when it breaks down.
6. Aesthetics	What does it look like?	How the product appeals to the senses — its design, color, shape, texture. This is subjective but important.
7. Features	What does the product do (beyond basics)?	Additional functions beyond the primary purpose. For a phone: camera, GPS, face recognition are features beyond basic calling.
8. Perceived Quality	What is the reputation of the company?	The customer's overall impression based on brand image, advertising, word-of-mouth. A Toyota is "perceived" as more reliable than many competitors even before testing it.

12. Dimensions of Service Quality — RATER Model

For services (unlike products), quality is measured using a different framework. The five dimensions of service quality form the **RATER** model:

Dimension	Meaning	Example
R — Reliability	Ability to perform the service consistently and accurately	A bank always processes transactions correctly
A — Assurance	Knowledge and courtesy of employees; ability to inspire trust	A doctor who explains the diagnosis clearly and confidently
T — Tangibles	Physical facilities, equipment, appearance of personnel	A clean, well-furnished hotel lobby with neatly dressed staff
E — Empathy	Caring, individualized attention to customers	A customer service agent who listens to your specific problem
R — Responsiveness	Willingness to help and provide prompt service	A restaurant that seats you quickly and takes orders promptly

> **Remember:** Product quality = Garvin's 8 Dimensions. Service quality = RATER model. Both are equally important in TQM, which covers products AND services.

13. Quality Definitions by Pioneers

Different quality gurus have defined quality from different perspectives. These definitions are frequently asked in exams:

Joseph Juran: "Quality is **fitness for use**."
Meaning: Quality of a product or service is the fitness of that product or service for meeting its intended use as required by the customer. It's not about luxury — a simple pen that writes well is a quality pen if writing is what you need.

Edward Deming: "Quality is **conformance to specifications**."
Meaning: A product has quality if it matches what was specified in the design. If the blueprint says a shaft should be 10mm diameter, then a shaft of 10mm is quality; 10.5mm is not.

Philip Crosby: "Quality is **conformance to requirements**."
Meaning: Similar to Deming but broader — quality is meeting all stated requirements, not just technical specs but also customer needs and regulatory requirements.

Feigenbaum: "Quality is **what the customer says it is**."
Meaning: The customer is the final judge of quality. No matter how good you think your product is, if the customer isn't satisfied, it's not quality.

ISO Definition: "The totality of features and characteristics of a product or service that bear on its ability to satisfy **stated or implied needs** of the customers."
Meaning: Quality includes both what the customer explicitly asks for AND what they implicitly expect.

Quality System (ISO): "A quality system is the agreed-upon company-wide and plant-wide operating work structure, documented in effective, integrated, technical and managerial procedures for guiding the coordinated actions of people, machines, and information of the company in the best and most practical ways to assure customer quality satisfaction and economical costs of quality."

Taguchi's View: "Quality is **inversely proportional to variations**."
Meaning: The less variation in your process/product, the higher the quality. This is the foundation of Six Sigma thinking.

14. Quality Gurus & Their Key Contributions

Five major quality gurus shaped modern quality management. Each had a unique philosophy:

Guru	Key Contributions	Core Philosophy
W. Edwards Deming	• Process orientation • Use of statistics • Drive out fear • Reduction of variation	Quality problems are mostly management problems. Use data and statistical methods. Create a fearless environment where workers can report problems.
Joseph Juran	• Management involvement • Quality planning • Quality control • Quality improvement	The "Juran Trilogy" — Quality Planning, Quality Control, and Quality Improvement form a continuous cycle. Management must be actively involved.
Armand Feigenbaum	• Total Quality System • Design quality in • Customer orientation	Coined "Total Quality Control" — quality must be built in at every stage, not just inspected at the end. The customer defines quality.
Kaoru Ishikawa	• Use of statistics • Quality circles • Involvement of employees	Democratized quality — every worker should participate. Created the Ishikawa (fishbone) diagram. Championed Quality Circles in Japan.
Philip Crosby	• Zero defects • Cost of quality • Hidden factory • Slogans	"Quality is free" — the cost of prevention is always less than the cost of correction. The "hidden factory" refers to the rework and scrap that consumes resources without adding value.

15. Three Aspects of Quality — Design, Conformance & Performance

Quality can be understood through three interconnected aspects that form a triangular relationship:

> **The Quality Triangle:**
> Quality of Design ←→ Quality of Conformance ←→ Quality of Performance
> (All three are linked — weakness in any one affects the others)

15.1 Quality of Design (Consumer's Perspective)

The product must be designed to meet the requirements of the customer. The product must be designed **right the first time and every time**. While designing, all aspects of customer expectations must be incorporated.

Factors to consider while designing:

- Type of product
- Cost
- Profit policy of the company
- Demand
- Availability of parts

15.2 Quality of Conformance (Manufacturer's Perspective)

The product must be manufactured **exactly as designed**. The activities at this stage include:

- Defect finding
- Defect prevention
- Defect analysis
- Rectification

Difficulties encountered at the manufacturing stage must be conveyed to designers for modification. **Two-way communication** between designer and manufacturer improves product quality.

15.3 Quality of Performance (Consumer's Perspective)

The product must function **as per the expectations of the customer**. Two-way communication between designers and customers is the key to having a quality product.

> **Analysis Point:** These three aspects show that quality is a chain: if the design is flawed (even if manufacturing is perfect), the product will fail. If the design is great but manufacturing introduces defects, quality suffers. And ultimately, the customer's experience of performance is what matters. TQM requires excellence in ALL three.

16. Evolution of Quality — Historical Timeline

Quality management has evolved through several distinct stages over more than a century:

Era	Period	What Happened
Operator Quality	~1890	Individual workers (operators) were responsible for their own quality. Each craftsman inspected his own work.
Foreman Quality	~1900	As factories grew, the foreman (supervisor) took over inspection responsibility. Workers focused on production; the foreman checked quality.
Inspection Era	~1918	Dedicated inspectors examined finished products. The focus was simply to **accept or reject** products based on specifications. No effort was made on defect prevention.
Statistical Quality Control (SQC)	~1924–1937	**Shewhart** (1924) created the first modern control chart. **Deming, Dodge, and Romig** developed the theory of Statistical Process Control (SPC). However, these techniques saw little use until the late 1940s.
Total Quality Control	~1950s–1960s	Quality management practices developed rapidly in **Japanese plants** and became a major theme in Japanese management philosophy. By 1960, quality control had become a **national preoccupation** in Japan.
"Total Quality" Coined	1969	**Feigenbaum** presented a paper using the term "total quality" for the first time, covering planning, organization, and management responsibility. **Ishikawa** explained how "total quality control" in Japan meant "company-wide quality control" — involving ALL employees from top management to workers.
TQM Era	1980–present	**Total Quality Management (TQM)** came into existence in 1980 by the western world. TQM became a holistic framework addressing overall organizational performance, continuous improvement, and customer/business results.

Analysis Point: The evolution shows a clear pattern: quality moved from being an **individual responsibility → specialized function → statistical science → organization-wide philosophy**. Japan adopted quality concepts faster than the West (1950s vs 1980s), which gave Japan a 30-year head start, explaining why Japan consistently tops quality rankings today.

17. Responsibility Towards Quality — 8 Stages of Industrial Cycle

Quality is not the responsibility of one department. The **8 stages of the industrial cycle** show how every function contributes:

#	Stage	Responsibility
1	**Marketing**	Determines customer and market quality requirements
2	**Engineering**	Translates requirements into quality-oriented designs
3	**Purchasing**	Procures quality materials and components
4	**Manufacturing Engineering**	Develops quality-oriented processes and procedures
5	**Manufacturing**	Transforms designs and materials into quality products
6	**Inspection & Test**	Determines quality of conformance
7	**Shipping**	Preserves and protects quality of product during transit
8	**Field Service & Installation**	Reports quality of performance back to the company

> **Key Insight:** This is a **closed loop** — Field Service feeds information back to Marketing, which starts the cycle again. Quality is a continuous cycle, not a one-time activity.

18. Quality Control — Definition, Importance & Benefits

18.1 What is Quality Control?

Quality Control (QC) is the **inspection, analysis, and action** applied to a portion of the product in a manufacturing operation to estimate overall quality of the

product and determine what changes must be made to achieve or maintain the required level of quality.

Quality control of a product can be viewed as a system which ensures:

- **Proper Planning**
- **Right Design**
- **Proper Equipment**
- **Proper Inspection**
- **Corrective Action**

18.2 Importance of Quality Control

- Quality is vital in **all areas** of business, including product development and production
- Cost of quality is ultimately **reduced** by investing money up front in quality design and development
- Typical costs of poor quality include: downtime, repair costs, rework, and employee turnover

18.3 Benefits of Quality Control

A well-established, committed quality system will deliver:

- Improvement in the quality of product
- Higher productivity
- Cost reduction
- Continuous improvement in quality of product

19. Stages of Quality Improvement

Quality improvement in an organization follows three progressive stages:

Stage	Description	What Happens
1. Commitment Stage	Introduction & Initial Implementation	The organization decides to adopt quality practices. Early efforts begin. Non-conformities (defects) are initially high but begin to decrease.
2. Consolidation Stage	Stabilization	Quality practices become routine. Non-conformities decrease steadily. Processes become more predictable and stable.
3. Maturity Stage	Continuous Quality Improvement	Quality is embedded in the culture. Non-conformities are minimal and continue to reduce. The organization achieves world-class quality levels.

> **Analysis Point:** The transition from Commitment to Maturity takes years, not months. Many organizations give up during the Commitment stage when they don't see immediate results. The key is persistence — the biggest quality improvements come during the Consolidation and Maturity stages.

20. Customer Expectations & the 100% Quality Target

What do customers really expect?

> **Customer Expectations:**
>
> - Deliver the customer what they exactly want — **First Time and Every Time**
> - **Nothing Less, Nothing More**
> - Quality Target = **100%**

Acceptable Quality = 99%? That sounds good, right? Think again:

Consider a **100 CM ruler** — can you accept a ruler that is 99 CM long? In many applications, even 1% defect rate is unacceptable. In aviation, healthcare, or nuclear industries, 99% quality means catastrophic failures.

> **Analysis Point:** The 100 CM ruler example is powerful. If 99% quality were acceptable:
> • 20,000 wrong prescriptions per year in a hospital
> • 2 unsafe landings at a major airport per day
> • 500 incorrect surgical operations per week
> This is why TQM targets **zero defects**, not "acceptable" defect levels.

Why TQM? — Real-World Business Cases

- **Ford Motor Company** had operating losses of **$3.3 Billion** between 1980 and 1982 — primarily due to quality issues
- **Xerox** market share dropped from **93% in 1971 to 40% in 1981** — Japanese competitors offered better quality copiers at lower prices
- Attention to quality was seen as the only way to **combat competition**

21. What is TQM? — Definitions & Meaning

21.1 Breaking Down TQM

Word	Meaning
Total	Made up of the whole — involves everybody in the organization
Quality	Degree of excellence a product or service provides
Management	Act, art, or manner of planning, controlling, directing

> **Therefore: TQM is the art of managing the whole to achieve excellence.**

21.2 TQM Definitions

> **Definition 1:** TQM can be seen as a process, used to manage the change in environment that will ensure the company reaches the goal of TCI (Total Continuous Improvement).
>
> **Total** = involves everybody in the organization
> **Continuous** = forever
> **Improvement** = elimination of waste, reduction of variations, innovation

> **Definition 2 (ISO 8402/IS 13999):** TQM is a management approach of an organization, centered on the **participation of all its members** and aiming at **long-term success** through customer satisfaction and benefits to the members of organization and society.

21.3 What Does TQM Mean in Practice?

TQM means that the organization's **culture** is defined by and supports the constant attainment of **customer satisfaction** through an integrated system of tools, techniques, and training. This involves the **continuous improvement** of organizational processes, resulting in high-quality products and services.

21.4 The Goal of TQM

> **"Do the right things RIGHT the first time, & every time."**

21.5 Two Main Objectives of TQM

1. **Total client satisfaction** through quality products and services
2. **Continuous improvements** to processes, systems, people, suppliers, partners, products, and services

22. TQM Principles & Basic Tenets

22.1 Four Core Principles of TQM

1. **Customer Orientation** — The customer is the ultimate judge of quality
2. **Management-by-Fact** — Decisions must be based on data and facts, not opinions or gut feelings
3. **Employee Involvement** — Every person in the organization must participate in quality
4. **Continuous Improvement** — Quality is a journey, never a destination

22.2 Five Basic Tenets of TQM

1. **The customer makes the ultimate determination of quality** — Not the engineer, not the manager — the customer
2. **Top management must provide leadership and support for all quality initiatives** — Without management commitment, TQM fails
3. **Preventing variability is the key to producing high quality** — Based on Taguchi's philosophy; reduce variation to improve quality
4. **Quality goals are a moving target, requiring commitment toward continuous improvement** — What's considered "quality" today may not be tomorrow
5. **Improving quality requires the establishment of effective metrics** — We must speak with data and facts, not just opinions or gut feelings

> **Analysis Point:** Tenet #3 connects to Taguchi's definition (Quality ∝ 1/Variation). Tenet #5 connects to Deming's emphasis on statistics. The tenets show that TQM isn't just philosophy — it requires measurable, data-driven action.

23. Productivity and TQM

Traditional View

"Quality cannot be improved without significant losses in productivity."

The traditional belief was that quality and productivity are trade-offs — improving one hurts the other.

TQM View

"Improved quality leads to improved productivity."

TQM rejects the trade-off myth. When you reduce defects (improve quality), you eliminate rework, reduce waste, decrease inspection costs, and increase customer satisfaction — all of which boost productivity.

> **Analysis Point:** This is one of the most important paradigm shifts TQM introduced. The traditional view treated quality as a cost. TQM treats quality as an investment that pays for itself many times over.

24. The Three Aspects of TQM — Counting, Customers, Culture

Aspect	Description
Counting	Tools, techniques, and training in their use for analyzing, understanding, and solving quality problems. This is the technical side of TQM — statistics, control charts, root cause analysis, etc.
Customers	Quality for the customer as a driving force and central concern. Everything the organization does should ultimately serve the customer.
Culture	Shared values and beliefs, expressed by leaders, that define and support quality. This is the human side — leadership, teamwork, empowerment, communication.

> **Key Insight:** All three are equally important. You can't have TQM with just tools (Counting) but no customer focus. You can't have customer focus without the right organizational culture. And culture without measurement tools is just wishful thinking.

TQM and Continuous Improvement

- TQM is the management process used to make **continuous improvements to all functions**

- TQM represents an ongoing, continuous commitment to improvement
- The foundation of total quality is a **management philosophy** that supports meeting **customer requirements** through **continuous improvement**

25. Continuous Improvement vs Traditional Approach

Traditional Approach	Continuous Improvement (TQM) Approach
Market-share focus	Customer focus
Individuals	Cross-functional teams
Focus on "who" and "why" (blame)	Focus on "what" and "how" (solutions)
Short-term focus	Long-term focus
Status quo focus	Continuous improvement
Product focus	Process improvement focus
Innovation (big jumps)	Incremental improvements (Kaizen)
Fire fighting (reactive)	Problem solving (proactive)

Analysis Point: The most important shift is from "who and why" to "what and how." Traditional management blames people when things go wrong. TQM focuses on fixing the process — because Deming taught us that 85% of quality problems are due to the system, not the workers.

26. The TQM System — Objectives, Principles & Elements

The TQM System can be visualized as a three-level hierarchy:

Level	Components	Description
Objective	Continuous Improvement	The overarching goal — never stop improving
Principles	Customer Focus, Process Improvement, Total Involvement	The three pillars that support the objective
Elements	Leadership, Education & Training, Supportive Structure, Communications, Reward & Recognition, Measurement	The practical building blocks that make TQM work in daily operations

> **Remember the hierarchy:** Elements support Principles, and Principles drive the Objective. Without strong Elements (like leadership and training), the Principles fail, and the Objective of continuous improvement cannot be achieved.

27. How to Implement TQM

There is no single way to implement TQM. Different approaches are possible, and in practice, interaction between them is needed:

Dimension	Option A	Option B
Style	**Soft** (cultural, people-focused)	**Hard** (tools, techniques, systems)
Approach	**Rhetoric** (philosophy, vision)	**Concrete Actions** (specific projects)
Scope	**Total Organization** (everyone at once)	**Pilot Project** (start small, then expand)
Focus	**General** (broad quality culture)	**Specific** (targeted improvements)
Direction	**Bottom Up** (workers drive change)	**Top Down** (management drives change)

Key Implementation Guidelines (from TQM Framework)

- Maintain at all times focus on the objective: to satisfy customers better than competitors do
- Turn employees from skill holders to disseminators — create learning
- Emphasize process more than product
- Get people to perform mundane activities with full quality focus
- Use technology efficiently
- Measure key parameters of all activities
- Build in continuous learning and improvement processes
- Get the workforce to manage itself
- Use customer and competitor data to drive all decisions
- Compensate for cultural weakness inherent in the organization
- Use facilitation to mediate structure changes
- Benchmark against the best — observe and copy their best practices
- Use scientific methods, not opinion or gut feel
- Deploy the workforce with sincerity and trust
- Use cross-functional teams
- Enable managers who go slow — don't filter people

28. Seven Steps of TQM

A structured approach to implementing TQM through seven sequential steps:

Step	Action	What to Do
1	Select a Theme	Choose a specific quality problem or improvement area to focus on
2	Grasp the Present System	Understand how the current process works — map it, measure it
3	Analyze the Present Situation	Use data and tools to identify root causes of quality problems
4	Set Countermeasures into Motion	Implement solutions to address the identified root causes
5	Determine Effectiveness	Measure whether the countermeasures actually improved quality
6	Use Standard Operating Procedures	Standardize the successful solutions so improvements are sustained
7	Plan for Future Action	Identify the next improvement cycle — repeat and improve continuously

Analysis Point: These seven steps are essentially a **PDCA (Plan-Do-Check-Act) cycle** in more detail. Steps 1–3 = Plan, Step 4 = Do, Step 5 = Check, Steps 6–7 = Act. This reinforces TQM's core principle of continuous improvement through structured cycles.

29. TQM Concepts & Techniques — Complete List

TQM employs a wide range of concepts and techniques. Here is the complete list from the lecture:

Core Concepts

- Continuous Improvement
- Customer Orientation
- Employee Involvement
- Management by Facts
- Process Control
- Transparency
- 5S
- 7 Management Techniques
- SPC Tools & Techniques
- Design of Experiments
- Six Sigma

Supporting Techniques

- Award Systems
- Benchmarking
- Complaints Handling
- Customer Surveys
- Empowerment
- Just-in-Time Management
- Quality Circles
- Quality Awards
- Quality Audits
- Quality Function Deployment (QFD)
- Quality Policy Deployment
- Reward Systems
- Top Management Commitment

30. 5S Concept — The First Step Towards TQM

5S is a Japanese workplace organization methodology that serves as the **foundation for implementing TQM**. It creates a clean, organized, and disciplined work environment — which is a prerequisite for quality production.

S#	Japanese	English	Description
1S	**Seiri**	**Sort**	Sort out unnecessary items in the work place and **discard them**. Keep only what is needed. Remove clutter, old tools, broken equipment, and unnecessary paperwork. Ask: "Is this needed here?"
2S	**Seiton**	**Set in Order**	Arrange necessary items in good order so that they can be **easily picked for use**. The principle is: **"A place for everything and everything in its place."** Labels, shadow boards, and designated storage areas help.
3S	**Seiso**	**Shine / Clean**	Clean your work place **completely** so that there is no dust on floor, machines, and equipment. Cleaning is also a form of inspection — you notice problems (leaks, cracks, wear) while cleaning.
4S	**Seiketsu**	**Standardize**	Maintain **high standards** of housekeeping and work place organization **at all times**. Create checklists, schedules, visual controls, and standard procedures so that 1S, 2S, and 3S are maintained consistently.
5S	**Shitsuke**	**Sustain / Self-Discipline**	Train people to follow good housekeeping disciplines **autonomously**. The goal is that employees maintain 5S without being told — it becomes a habit, part of the culture.

Analysis Point: 5S is called the "first step towards TQM" because:
1. It's **simple to understand** and implement — doesn't require complex statistical knowledge
2. It creates **visible, immediate results** — motivating employees to embrace further quality initiatives
3. It builds **discipline** — the same discipline needed for more advanced TQM practices
4. It **reduces waste** — directly supporting lean manufacturing and TQM goals
5. It's the foundation — you can't run complex quality systems in a messy, disorganized workplace

> **Exam Tip:** Remember the 5S in order: **S**eiri → **S**eiton → **S**eiso → **S**eiketsu → **S**hitsuke (Sort → Set in Order → Shine → Standardize → Sustain). The first 3S are actions, the 4th is maintaining, and the 5th is culture.

Chapter Summary — Key Takeaways

1. India's Quality Gap: India ranks last among major nations in price-to-quality, customer orientation, and TQM practice. Despite having the 3rd largest pool of scientists/engineers, quality of engineers ranks 52nd out of 53.

2. Quality is Everyone's Job: Not just the quality department — every function from marketing to shipping contributes to quality (8 stages of industrial cycle).

3. Key Definitions: Juran (fitness for use), Deming (conformance to specifications), Crosby (conformance to requirements), Feigenbaum (what the customer says), Taguchi (inversely proportional to variation).

4. Product vs Service Quality: Products = Garvin's 8 Dimensions (Performance, Conformance, Reliability, Durability, Serviceability, Aesthetics, Features, Perceived Quality). Services = RATER (Reliability, Assurance, Tangibles, Empathy, Responsiveness).

5. TQM = Total + Quality + Management: The art of managing the whole to achieve excellence. Goal: "Do the right things right, the first time and every time."

6. Four TQM Principles: Customer Orientation, Management-by-Fact, Employee Involvement, Continuous Improvement.

7. Quality Evolution: Operator (1890) → Foreman (1900) → Inspection (1918) → SQC (1924) → TQC (1950s) → TQM (1980s). Japan was 30 years ahead of the West.

8. TQM vs Traditional: TQM focuses on customers (not market share), processes (not products), "what/how" (not "who/why"), long-term (not short-term), and problem-solving (not fire-fighting).

9. 5S is the Foundation: Seiri (Sort) → Seiton (Set in Order) → Seiso (Shine) → Seiketsu (Standardize) → Shitsuke (Sustain). The simplest and first step towards TQM.

10. Improved Quality = Improved Productivity: TQM rejects the myth that quality and productivity are trade-offs. Better quality → less rework → more efficiency → higher productivity.